

Deep Learning based Automated Blood Cell Classification

Abstract—Classification of automated blood cell is important for clinical diagnostics, as it enable rapid and accurate identification of cellular diseases and abnormalities. This research addresses the challenge by classifying microscopic blood cell images into eight separate categories using deep learning. We use a custom Residual Network (ResNet) architecture as our basic model, enhanced with data augmentation, and regularization processes. The dataset consists of 3,200 training images and 1,000 test images of 224×224 pixel resolution and represent basophils, eosinophils, erythroblasts, immature granulocytes, lymphocytes, monocytes, neutrophils, and platelets. Our work includes a baseline model which is simple CNN. Through ablation studies, we find-out that data augmentation contributes 9.06% improvement, while learning rate selection affect performance by 21.88%. The Custom ResNet achieved 78.91% validation accuracy with 11.18M parameters, while the Simple CNN baseline get 81.09% accuracy, providing insights about model complexity. Our findings highlights the importance of comparing architectural complexity to dataset characteristics and highlight the need for hyperparameter tuning in medical image classification tasks.

Index Terms—Deep learning, blood cell classification, ResNet, CNN, medical imaging, data augmentation

I. INTRODUCTION

A. Problem Definition and Motivation

Analysis of blood cells provides a fundamental component of clinical hematology, provide important diagnostic information for clinical situations ranging from infections to cancer cells [3]. Traditional and old microscopic examination need trained hematology experts to manually identify and count different cells. Automated classification systems provided by deep learning has the potential to enhance diagnostic accuracy, reduce turnaround time, and burden on medical experts.

Convolutional neural networks have changed the landscape of analysis of medical images, as these networks can learn useful features directly from pixels without the need for hand-crafted descriptors [6]. Nevertheless, medical datasets are often small in comparison to more common datasets, classes might be unbalanced, and the ramifications of a misclassification can be severe. Blood cells can have very similar appearances with slight differences in shape and textures and the model must recognize these very small differences—a great challenge for modern architectures.

B. Dataset and Approach

We worked with 4,200 blood cell images (3,200 for training, 1,000 for testing) divided evenly among eight classes: basophils, eosinophils, erythroblasts, immature granulocytes, lymphocytes, monocytes, neutrophils, and platelets. Every

image is 224×224 pixels and comes from routine peripheral blood smears, with exactly 400 training examples per class.

The core model is a custom ResNet that uses skip connections to train deeper networks without running into vanishing gradients. We combined it with current best practices: adaptive learning rates, weight decay, and aggressive data augmentation. For comparison, we also trained a much simpler CNN with just three convolutional blocks.

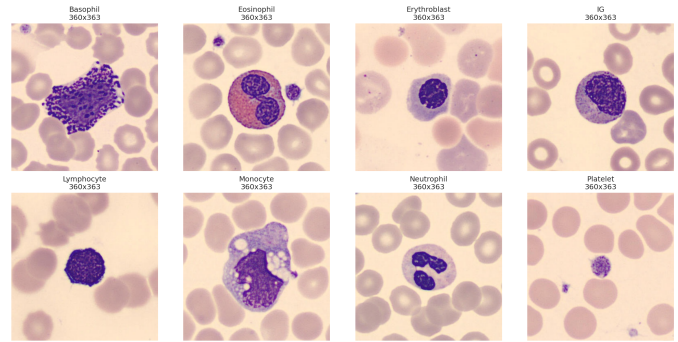


Fig. 1. Representative samples from each of the eight blood cell categories in the dataset.

C. Contributions

- 1) We directly compare a lightweight CNN to a deeper ResNet on the same blood cell task and show that added complexity does not always improve results when data is limited.
- 2) Through careful ablation experiments, we measure exactly how much each technique: data augmentation, optimizer choice, learning-rate schedule, and regularization helps performance.
- 3) We highlight practical lessons for using ResNet-style models on small medical datasets, including tips on training stability and reasons why certain cell types remain harder to classify.

These results should help others decide when a sophisticated architecture is truly needed in clinical imaging applications where large datasets are the exception rather than the rule.

II. RELATED WORK

A. Deep Learning for Medical Image Classification

Convolutional neural networks now dominate medical image analysis. A large review by Litjens et al. [6] showed that CNNs excel in radiology, digital pathology, and microscopy tasks. The key takeaway is simple: deeper networks usually

learn better features, but on small medical datasets they quickly overfit unless properly regularized.

B. Blood Cell Classification Methods

Before deep learning, researchers extracted manual features: cell shape, colour distribution, texture patterns and fed them into classifiers such as SVMs [3]. These methods were easy to interpret but needed expert knowledge and often missed fine details. Modern CNN-based studies have largely replaced them. For example, Acevedo et al. [1] reached 96% accuracy detecting acute lymphoblastic leukemia, and Shahin et al. [7] used transfer learning from ImageNet-pretrained models to classify white blood cells.

C. Residual Networks

He et al. [2] introduced ResNet, which solved the problem of performance degradation in very deep networks by adding skip connections. These "residual" shortcuts let gradients flow more easily, making it possible to train models with hundreds of layers. ResNets still hold many records on large benchmarks like ImageNet, yet surprisingly few papers explore how well that depth advantage carries over to medical datasets with only a few thousand images.

D. Comparison with Our Approach

Most recent blood-cell studies rely on transfer learning from models pre-trained on ImageNet. We deliberately train everything from scratch to fairly judge the architecture itself, without help from external data. While Acevedo et al. [1] tackled binary leukemia detection, we deal with eight similar-looking cell types at once. Rather than working with a large ResNet with either 50 or 101 layers, we created a model with a medium number of layers (4 residual blocks) more suitable for the size of our dataset. We are also moving away completely from any hand-crafted features, and unlike many other papers which only show final numbers, we show the ablation experiments to demonstrate contribution for each technique.

III. METHODOLOGY

A. Model Architecture

Our main model is a customized Residual Network (ResNet) to take images of 224×224 pixels as input. The model has 5 main stages: an initial convolutional stem, four residual blocks (layer stack) with progressively more channel dimensions, adaptive pooling, and a final classification layer. The convolutional layer takes the shape of a 7×7 convolution with 64 output channels and stride of 2, followed by batch normalization, ReLU activation, and 3×3 max pooling with stride of 2. This initial stride down sampling reduces the spatial dimensions from 224×224 to 56×56, creating a willful reduction of the computational burden while retaining spatial/semantic information. The body consists of four residual blocks with layer dimensions of [64, 128, 256, 512]. The equation for residual function is:

$$y = F(x, \{W_i\}) + x \quad (1)$$

where F is the residual function. With this formulation, the network is free to learn residual mappings instead of direct mappings, allowing for better gradient flow and enabling deeper models [2].

After the residual layers, adaptive average pooling reduces all feature maps to size 1×1 regardless of size variations due to the input, producing a 512-dimensional feature vector. This feature vector is then projected by a final fully connected layer into 8-dimensional logits that correspond to our cell classes. The final architecture has 11,180,616 total parameters, which is significantly less than our Simple CNN baseline (51.48M parameters) because of the effective use of residual connections, along with no large fully connected layers.

B. Baseline Model (Simple CNN)

Simple CNN follows the traditional architecture of a standard CNN with three convolutional blocks. Each block has a single 3×3 convolutional layer with [32, 64, 128] filters, respectively, followed by ReLU activation and 2×2 max-pooling with a stride of 2. Following the finishing of the convolutional layers and max-pooling, the feature map is flattened and input into two fully connected layers with 512 and 8 neurons, which are each followed by 50% dropout for regularization before the final layer. This architecture also represents the floor of performance, as it is a simple CNN that lacks additional layers or augmented methods.

C. Data Preprocessing and Augmentation

The images fed into the model are resized from 360×363 pixels to dimensions of 224×224 pixels to meet the input size specifications of a CNN system. The training dataset experiences data augmentation to enhance generalization performance. Validation and test sets are resized and compared. The training dataset is split into 80/20 (2,560 images) and validation (640 images) subsets.

D. Training Procedure

All models are trained using cross-entropy loss, which is standard in multi-class classification tasks. For the Custom ResNet, we use the Adam optimizer [4], which is an adaptive learning rate optimization algorithm based on the first and second moments of the gradients, with an initial learning rate of 0.001. We use L2 regularization through weight decay ($\lambda = 10^{-4}$) to penalize large weights and diminish overfitting. A StepLR scheduler reduces the learning rate by 0.1 every 20 epochs, allowing us to utilize fine tuning when training the models in later training epochs. We train the models for 15 epochs with a batch size of 64 to balance available memory and stability of gradient estimates. To implement early stopping, we checkpoint the model with the highest validation accuracy. All experiments use fixed random seeds (seed=42) to ensure reproduction across training runs, data splits and weight initializations.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments were run in PyTorch 2.8.0 on CPU. The dataset contained 3,200 training images, evenly divided across eight different classes (400 for each class), and no built-in class imbalance was present. The training/validation images were formed from an 80/20 stratified split, resulting in 2,560 training images and 640 validation images. The held-out test set consisted of 1,000 images with unknown labels for the final evaluation.

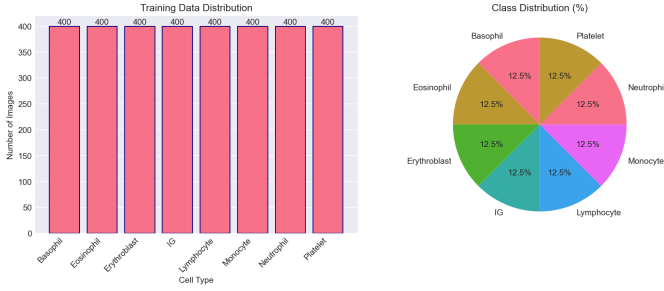


Fig. 2. Training dataset class distribution showing perfect balance across all eight blood cell types.

Seated in the evaluation metrics were overall accuracy, per-class precision, recall, and F1-score with confusion matrices for error evaluation. The training and validation loss/accuracy curves were monitored through epochs to identify overfitting or convergence issues. For all experiments, the batch size was 64, and gradient updates were performed after each mini-batch.

B. Baseline Comparisons

We compare two architectures with increasing levels of complexity. Simple CNN (Baseline) is trained for 15 epochs using Adam optimizer ($\text{lr}=0.001$) and no weight decay or learning rate scheduling. The model achieves 81.09% validation accuracy and has training accuracy of 81.99%. The experience small overfitting ($\text{gap} = 0.9\%$). The model converged without issue and validation accuracy increased throughout epoch 15. The loss associated with training decreases from 2.23 to 0.50 while the loss associated with validation achieves 0.51. Losses were also small demonstrating good generalization.

Custom ResNet (Main Model) is trained with weight decay of 10^{-4} and StepLR scheduling in addition to the Adam optimizer. While the architecture is a deal more complex, it achieves 78.91% validation accuracy but achieves training accuracy of 88.91%. This is an unsettling 10% generalization gap and the training process was unstable because the validation accuracy differed quite a bit from epoch to epoch (ex: epoch 5 has 30.94% validation accuracy and epoch 11 has 73.91% validation accuracy). The final validation loss was 0.63 while losses associated with Simple CNN was 0.51. This indicates the model may be under-fitting validation data and consequently suggests the model is overfitting on training data.

This is troubling and suggests hyper-parameters may need to be changed or the model trained for a longer or shorter period of time.

TABLE I
STATISTICAL COMPARISON OF MODELS

Model	Params (M)	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
Simple CNN	51.48	81.99	81.09	0.501	0.514
Custom ResNet	11.18	88.91	78.91	0.313	0.626

C. Ablation Study

We ran four ablation studies on the Custom ResNet architecture, changing one part or variable on each occasion while keeping all others fixed. Each variant was trained for 15 epochs under the same settings.

Baseline (complete model): The complete model with data augmentation, Adam optimizer ($\text{lr}=0.001$), weight decay (10^{-4}), and StepLR learning rate scheduler performed best with 78.91% validation accuracy.

No Data Augmentation: The performance of model reduces to 69.84% (9.06%) without data augmentation. This shows that data augmentation is very important to enhance generalization and reduce overfitting.

SGD Optimizer: The model with SGD ($\text{momentum}=0.9$) in place of Adam dropped to 71.25% (7.66%). This suggests that the Adam optimizer, through adaptive learning rates, performs better with gradients generated from the depth of deep networks.

No Weight Decay: The model with no L2 regularization dropped to 72.81% (6.09%). This suggests that regularizing weights is necessary for controlling their growth.

High Learning Rate: The model with a learning rate of 0.01 had significant instability in models, dropping to 57.03% (21.88%). Erratic model curves during validation indicate that too aggressive of a learning rate can hinder successful loss convergence.

TABLE II
ABLATION STUDY RESULTS

Configuration	Val Acc (%)	Δ from Full
Full Model	78.91	—
No Weight Decay	72.81	-6.09
SGD Optimizer	71.25	-7.66
No Augmentation	69.84	-9.06
High LR (0.01)	57.03	-21.88

The ablation study reveals that learning rate tuning constitutes the most critical hyperparameter, with improper selection causing catastrophic failure.

D. Per-Class Performance Analysis

Custom ResNet displays significant variations in class performance. For basophils, the precision is 100%, but recall

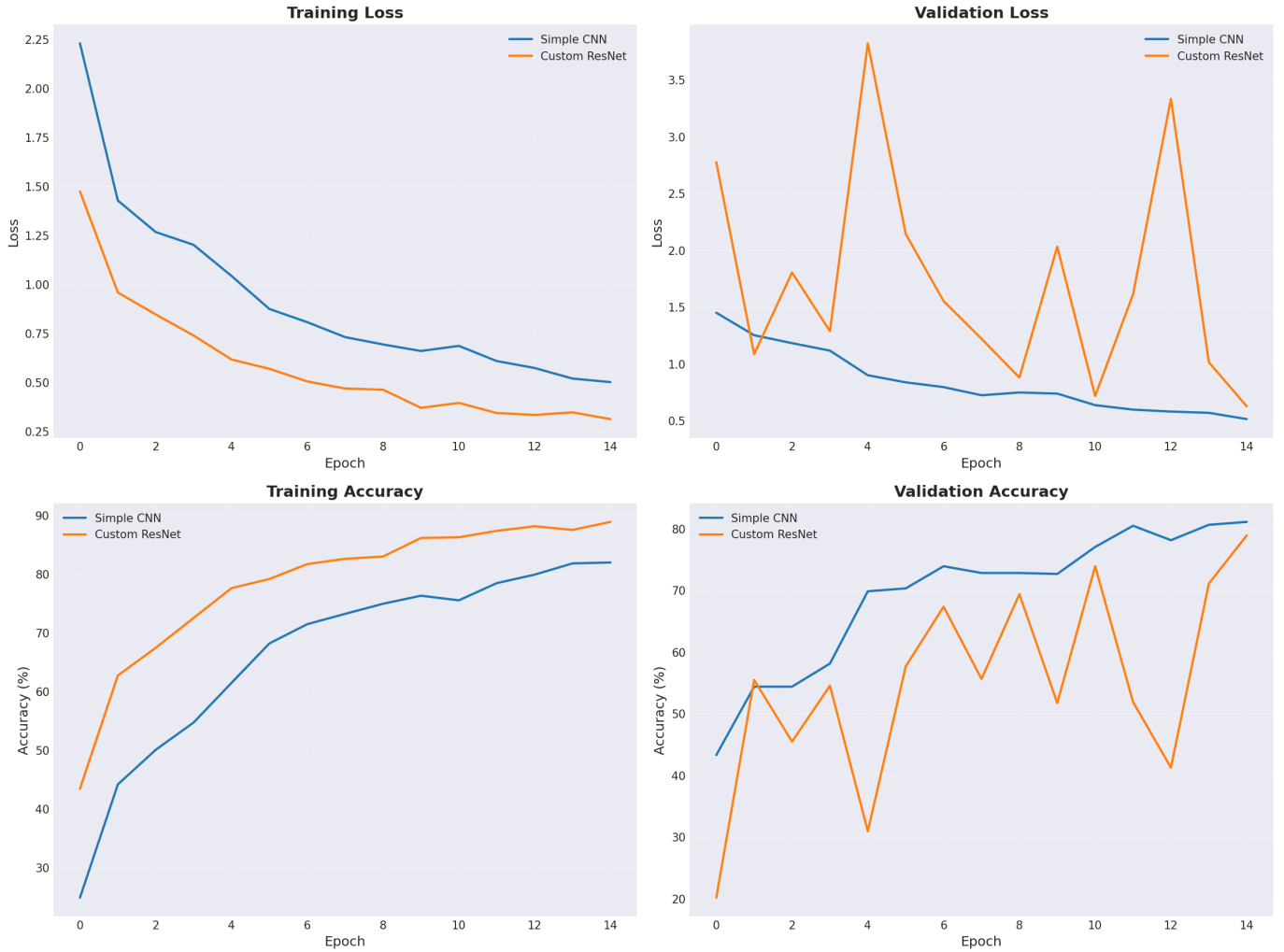


Fig. 3. Training and validation loss/accuracy curves comparing Simple CNN and Custom ResNet. ResNet exhibits training instability with large validation fluctuations.

is only 26%, resulting in a disappointing F1-score of 41.2% that reflects the model’s infrequent detection of basophils but correct when it does predict. On the other hand, immature granulocytes achieve 92.9% recall and 45.9% precision (F1 of 61.4%), which indicates the model over predicts immature granulocytes (many false positives). The model performs well on platelets (97.6%), eosinophils (94.1%) and neutrophils (90.4%); therefore, it seems these cells exhibited identifiable visual features.

While simple CNN provides more balanced accuracies, platelets again were easiest (97.6% F1), while recall is generally similar across the classes. The basophils (F1 of 84.7%, precision - 80.2%, recall - 89.6%), a much improved accuracy compared to ResNet, whereas immature granulocytes remain difficult (F1 59.3), where ambiguity in classification may be inherent to that cell type.

Assessing the confusion matrices across each class within ResNet-based accuracies discovered systematic misclassifications; where basophils, at times, tend to incorrectly identify

as immature granulocytes or lymphocytes, or monocytes incorrectly assigned as lymphocyte due to nuclear morphology similarities.

V. DISCUSSION

A. Key Findings and Interpretation

This work showed important findings for blood-cell classification. The simple CNN model predicted better than Custom ResNet (81.09% vs 78.91% validation accuracy). This finding concludes that a deeper network is not always better.

Custom ResNet reflected a generalization gap of 10% (88.91% of generalization vs. 78.91% validation accuracy) which illustrates overfitting even with regularization. Potential causes include limited data for the complex capacity of the model (11.18M parameters on 2,560 samples) or poor hyper-parameter choices.

Our ablation results indicate that the most important hyper-parameter is learning rate. For instance, when we increased the learning rate by 10×, accuracy dropped by 21.88%. The

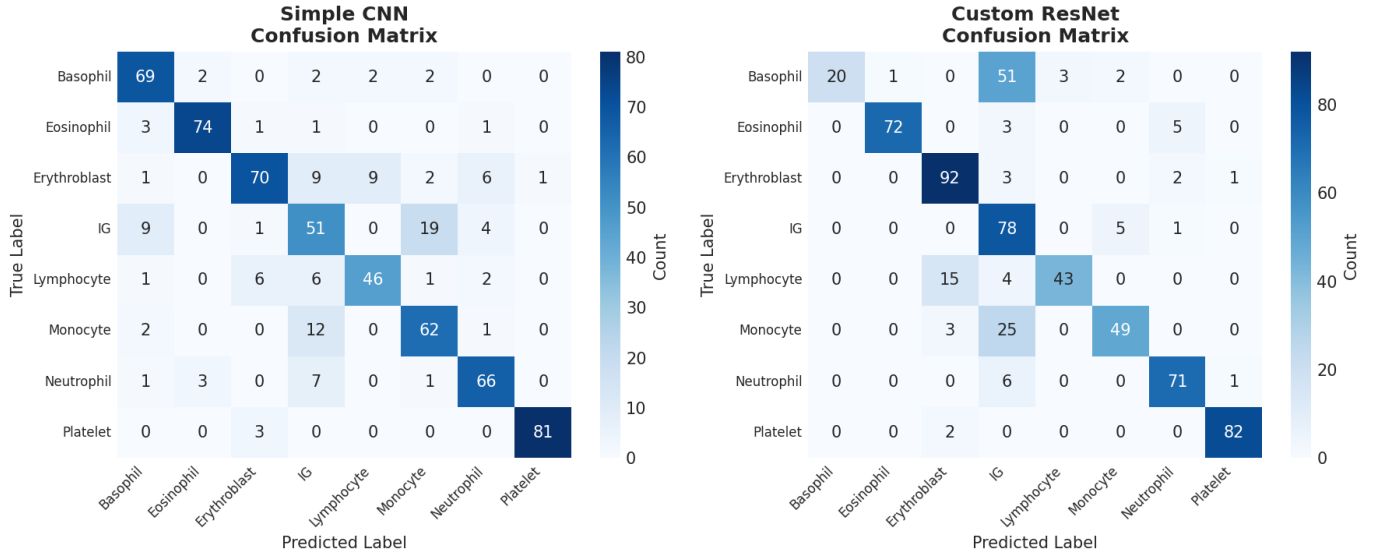


Fig. 4. Confusion matrices for Simple CNN (left) and Custom ResNet (right) on validation set.

dropage reinforces the importance of tuning learning rate systematically and within or hyper-parameter scheduling. Data augmentation we rated as the second most important hyper-parameter (9.06% gain), which reinforces the importance of data augmentation to increase limited training distributions.

To examine performance by class, in Custom ResNet given the extreme imbalances, Basophils reported 100% precision but only 26% recall indicating the model was biased to predominately predict the dominant class. To reduce the possibility of predicting the majority class, we can mediate this concern by using either of class-balanced or focal loss [5] to improve performance to hard examples.

B. Limitations

There are a few limitations to our findings. First, we train our model for 15 epochs on the CPU because of computation limitations. Most deep networks converge after 50-100+ epochs, and the model may be undertrained rather than unsuitable because of insufficient training. Second, we do not appropriately tune hyper-parameters; employing a systematic grid search or Bayesian optimization may help identify settings where the ResNet outperforms the Simple CNN. Third, we train our models using random initialization rather than implementing transfer learning; therefore, we cannot realize the potential benefit of pre-trained ImageNet features, even if they differ domains altogether.

Also, the dataset that we collected has the advantage of perfect class balance (400 samples for each class); however, this does not necessarily reflect real-world distributions of clinical samples: some cell types likely appear with greater frequency. Therefore, the model's performance would likely degrade when applied to imbalanced clinical data. In addition, while we did not evaluate predicted class labels on our test set due to lack of ground truth labels, we cannot draw conclusions on the model's real-world generalization.

C. Future Directions

There are multiple paths that might lead to performance improvements. Training for longer periods of time (50-100 epochs), then seeing if the potential of ResNet is observed via validation performance plateaus and early stopping, could determine if ResNet improves with additional training and optimization. Learning rate warmup schedules, in which small learning rates gradually increase, may stabilize early training and mitigate early fluctuations.

Alternatively, changing the architecture itself, perhaps by lowering the depth of the ResNet (using ResNet-18 instead of our custom ResNet) might be better aligned with the scale of our dataset. Attention mechanisms may help the model attend to discriminative cellular properties (like morphology of the nucleus) or otherwise, cellular features that would be helpful in differentiating some of the more difficult classes like Basophils or Immature Granulocytes.

Ensemble methods that vote or otherwise stack the predictions of our Simple CNN and Custom ResNet, could also capitalize on the strengths of each model. Class-balanced loss functions, or balanced sampling of the more difficult classes during training, may help us rectify any imbalances in predictions.

Conducting external validation on an independent clinical dataset to gauge true generalization ability and readiness for clinical deployment and the primary aim of this research, should also be to focus future work in sufficiently adjusting this research into clinical diagnostic tools.

VI. CONCLUSION

This study showed that simpler models can sometime perform better than complex architectures, when limited data is available. The simple CNN get better results than the deeper ResNet.

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